

# Design of a MOSFET Circuit Used in FOC Driver to Reduce the Loss and Improve the Efficiency

Shihao Han

Beijing University of Technology, Beijing, 100124, China

**Abstract.** Robot arms are being used in many fields at now, as the core of it, FOC plays the role in controlling function. This paper provides an overview of the specific function for MOSFET circuit used in FOC driver, which could reduce the loss of energy, improve the efficiency and extend the life of machine. Firstly, paper provides the three-phase inverter bridge circuit to produce eight states, which forms the SVPWM to control the electronic motor. Next, we do simulation in MATLAB to observe the specific motor states and the voltages in output. Finally, using different simulation environment to get the comparison with the expectation and raise the improvement of efficiency and power. This paper pays much attention to the introduction of three-phase inverter bridge and the SVPWM used in control motor.

**Keywords:** Robot-arms; MOSFET; SVPWM; FOC.

## 1. Introduction

With the development of robot arm used in medical, manufacturing, scientific research and other fields, as the main control core of robot arm, improve the efficiency of FOC has been raised all over the world. Moreover, to control the electronic motor (FOC control unit), how to design the three phase voltages wave produced by inverter bridge is vital, reduce the loss of energy and improve efficiency in the circuit is in the priority in designing.

FOC (field-oriented control, also known as vector frequency conversion, is currently the best choice for efficient control of brushless DC motors (BLDC) and permanent magnet synchronous motors (PMSM). The high switching frequency is obtained due to the application of high-efficient power switching components and powerful microprocessor. It allows to achieve good dynamical properties of current control loop, proper disturbance compensation and silent operation of servo-drive.[1] At present, it has gradually replaced the traditional control method in many applications, and has attracted much attention in the motion control industry. Except for the excellent algorithm of the FOC, the excellent driver design often determines the final effect of motor control. In the driver module, inverter bridge circuit made by MOSFET could be used as Generating a sine wave current to drive the motor, which is the main heat source of PCB. In this paper, the principle of three-phase inverter bridge and how it is used in the motor controlling will be explained as focus. In the field of motor control, in order to pursue the miniaturization of motor controller, parallel MOSFET is often used as the power output module. Although many characteristics of MOSFETs are easy to use in parallel, the overcurrent problem caused by the inconsistent switching process has not been fundamentally solved. In this paper, the current state of the power device used in SVPWM control algorithm is analyzed in detail. The problem of power device switch inconsistency is solved in general topology[2].

Although there are numerous designs of inverter bridge and the relationship between three-phase voltage and SVPWM, most of them faces the precision or complex problems. In the paper of chan WL, direct speed control that involves a simpler circuitry and reasonable speed control is proposed, but there is always a small steady-state error due to the absence of a proportional–integral controller.[3]. Although it has been proposed by the O.C. Kivanc and S.B. Ozturkrelation about between three-phase four switches and SVPWM, it is limited in some complex circuit. [16]

In this paper, the relationship between three-phase inverter bridge and SVPWM would be explained briefly, the SVPWM used in specific environment and its reliability, precision and power

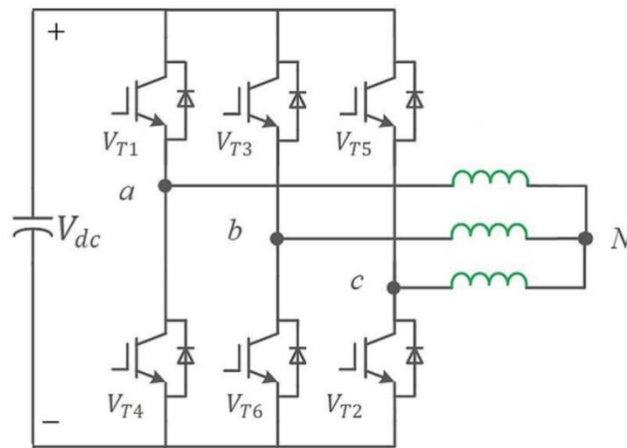
under this circumstance would be compared in both of calculation and modulation. Additionally, the performance of three-phase six switches circuit should be compared with other inverter bridges.

A. Introduction of the relationship between different states in circuit and sinusoidal wave in SVPWM and three-phase inverter bridge SVPWM.

SVPWM (Space Vector Pulse Width Modulation) is a controlling method, which is a pulse width modulated wave generated by a specific switching mode composed of six power switching elements of a three-phase power inverter, which can make the output current waveform as close to an ideal sine waveform as possible.[2]

The same SVPWM control algorithm is used to control the three-phase inverter power IGBT switch. Compared with the traditional sinusoidal PWM inverter, the DC bus voltage utilization rate of SVPWM voltage source inverter is increased by 15%, and the output harmonic distortion is low [12] - [14]. The control strategy of inverter is V/ F control method based on space vector modulation technology. The model of SVPWM technology is simulated by MATLAB Simulink program.

B. Three-phase inverter six switches circuit



**Fig 1.** Three-phase inverter six switches circuit

In order to achieve the purpose of controlling the motor, the motor drive system must be able to provide the motor with a voltage whose frequency and amplitude can be changed. The frequency and amplitude of the industrial power supply output voltage are fixed. We can first turn the voltage output by the industrial power supply into a DC voltage source,  $V_{dc}$ , which is what you can see in Figure 1, this step is what we called rectification. This DC voltage source can be transformed into a voltage with variable frequency and amplitude through the three-phase inverter in Figure 1, and this step is called inversion. The inverter in Figure 1 is connected to the three stator phase windings of the motor, and the inverter has three bridge arms, a, b, c. So it is called a three-phase inverter. There are two switches on each bridge arm. For example, in phase we have and two switches,  $V_{T1}$  and  $V_{T4}$ , which control the turn-on and turn-off of the upper and lower half bridge arms of the phase, respectively. The point N in Figure 1 represents the neutral point of the three-phase winding of the motor.

In the three-phase inverter of Figure 1, the two switch states on each phase leg are complementary. If the upper switch of the phase is on, then the lower switch must be off. We can define a switch function first.

$$S_x(x=a,b,c)$$

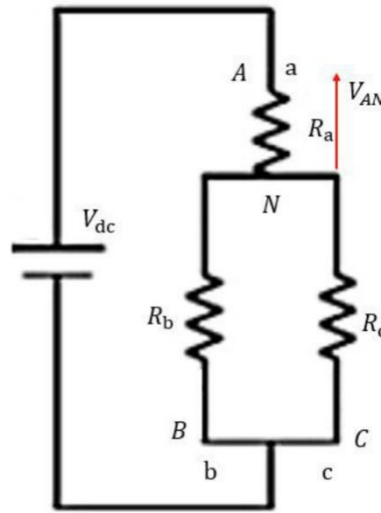
For example,  $S_a$  represents the switch in a's upper switch is on, the lower switch is off,  $S_a=0$  has the opposite meaning. If in Figure 1, the Inverter  $V_{T1}, V_{T6}, V_{T2}$  switches are on,  $V_{T4}, V_{T4}, V_{T5}$  switches are off, the state corresponds to the function is

$$S_a = 1, S_b = 0, S_c = 0.$$

These three switch function values can be represented by a vector (1 0 0), and of course they can also be regarded as a three-digit binary number. The inverter has three phases and they can be

represented by three switching functions, and the values of these three switching functions can be combined in a variety of  $2 \times 2 \times 2 = 8$  forms.

The motor winding connection diagram corresponding to this state is as follows



**Fig 2.** Motor winding connection diagram

According to figure2, A,B,C represents three phase windings of motor, a,b,c represents the connection point between the three phases of the three-phase inverter and the motor winding in Figure 1, and they are in the middle of the three phases.  $V_{AN} = -V_{NA}$  is the phase voltage of the motor phase winding (that is, the phase-to-neutral voltage of the motor winding), where  $V_{BN}$  and  $V_{CN}$  has the same meaning. The SVPWM algorithm assumes that the motor has three-phase balanced windings, which means that the phase resistances of all three phases of the motor are equal.[3] Phase resistance  $R_a$  and  $R_b$  is parallel, their total resistance is half of the phase resistance  $R_a$  (three resistances are equal).

According to the series voltage division theorem, we have

$$V_{AN} = 2V_{NB}$$

In the parallel circuit, we have

$$V_{NB} = V_{NC}$$

The total voltage drop can be expressed as

$$V_{dc} = V_{AN} + V_{NB}$$

So far, the three-phase voltage of the motor can be represented by the DC voltage of the inverter in Figure 1

$$V_{AN} = \frac{2}{3}V_{dc}$$

$$V_{NB} = V_{NC} = \frac{1}{3}V_{dc}$$

$$V_{BN} = V_{CN} = -\frac{1}{3}V_{dc}$$

The above discussion is only one state (1 0 0) of several switches of the inverter in Figure 1, we have 7 other states (0 0 1), (0 1 0), (0 1 1), (1 0 1), (1 1 0), (0 0 0), (1 1 1). The three-phase voltages of the motor corresponding to the remaining 7 states can also be represented by DC voltages, and they are listed in Table 1.

Converting binary (1 0 0) to decimal is the number 4, so the subscript 4 represents this state. The three phase windings of the motor are placed  $120^\circ$  apart from each other in space, then the phase voltages on the three phase windings of the motor are combined into a space voltage vector in space.  $V_4$  is the inverter three-phase switching function  $S_a = 1, S_b = 0, S_c = 0$ 's space voltage vector of the combined phase voltages on the three phase windings of the motor.

**Table 1.** State table for phase voltage

| Sa | Sb | Sc | Vector | Phase voltage        |                      |                      |
|----|----|----|--------|----------------------|----------------------|----------------------|
|    |    |    |        | $V_{AN}$             | $V_{BN}$             | $V_{CN}$             |
| 0  | 0  | 0  | $V_0$  | 0                    | 0                    | 0                    |
| 1  | 0  | 0  | $V_4$  | $\frac{2}{3}V_{dc}$  | $-\frac{1}{3}V_{dc}$ | $-\frac{1}{3}V_{dc}$ |
| 1  | 1  | 0  | $V_6$  | $-\frac{1}{3}V_{dc}$ | $\frac{2}{3}V_{dc}$  | $-\frac{1}{3}V_{dc}$ |
| 0  | 1  | 0  | $V_2$  | $-\frac{1}{3}V_{dc}$ | $\frac{2}{3}V_{dc}$  | $-\frac{1}{3}V_{dc}$ |
| 0  | 1  | 1  | $V_3$  | $-\frac{2}{3}V_{dc}$ | $\frac{1}{3}V_{dc}$  | $\frac{1}{3}V_{dc}$  |
| 0  | 0  | 1  | $V_1$  | $-\frac{1}{3}V_{dc}$ | $-\frac{1}{3}V_{dc}$ | $\frac{2}{3}V_{dc}$  |
| 1  | 0  | 1  | $V_5$  | $\frac{1}{3}V_{dc}$  | $-\frac{2}{3}V_{dc}$ | $\frac{1}{3}V_{dc}$  |
| 1  | 1  | 1  | $V_7$  | 0                    | 0                    | 0                    |

The calculation of  $V_1, V_2, V_3, V_4, V_5, V_6, V_7$  is as follow

$$\begin{aligned}
 V_4 &= \frac{2}{3} \left( \frac{2}{3}V_{dc} - \frac{1}{3}V_{dc} * e^{j\frac{2\pi}{3}} - \frac{1}{3}V_{dc} * e^{j\frac{4\pi}{3}} \right) = \frac{2}{3}V_{dc} \\
 V_6 &= \frac{2}{3} \left( \frac{1}{3}V_{dc} + \frac{1}{3}V_{dc} * e^{j\frac{2\pi}{3}} - \frac{2}{3}V_{dc} * e^{j\frac{4\pi}{3}} \right) = \frac{2}{3}V_{dc} \left( \frac{1}{2} + j\frac{\sqrt{3}}{2} \right) = \frac{2}{3}V_{dc} * e^{j\frac{\pi}{3}} \\
 V_2 &= \frac{2}{3} \left( -\frac{1}{3}V_{dc} + \frac{2}{3}V_{dc} * e^{j\frac{2\pi}{3}} - \frac{1}{3}V_{dc} * e^{j\frac{4\pi}{3}} \right) = \frac{2}{3}V_{dc} \left( -\frac{1}{2} + j\frac{\sqrt{3}}{2} \right) = \frac{2}{3}V_{dc} * e^{j\frac{2\pi}{3}} \\
 V_3 &= \frac{2}{3} \left( -\frac{2}{3}V_{dc} + \frac{1}{3}V_{dc} * e^{j\frac{2\pi}{3}} + \frac{1}{3}V_{dc} * e^{j\frac{4\pi}{3}} \right) = -\frac{2}{3}V_{dc} \\
 V_1 &= \frac{2}{3} \left( -\frac{1}{3}V_{dc} - \frac{1}{3}V_{dc} * e^{j\frac{2\pi}{3}} + \frac{2}{3}V_{dc} * e^{j\frac{4\pi}{3}} \right) = \frac{2}{3}V_{dc} e^{j\frac{4\pi}{3}} \\
 V_5 &= \frac{2}{3} \left( \frac{1}{3}V_{dc} - \frac{2}{3}V_{dc} * e^{j\frac{2\pi}{3}} + \frac{1}{3}V_{dc} * e^{j\frac{4\pi}{3}} \right) = \frac{2}{3}V_{dc} e^{j\frac{5\pi}{3}} \\
 V_0 &= V_7 = 0
 \end{aligned} \tag{1}$$

Formula (1) for 8 motor states voltages.

Three phase motor stators

The stator has A,B,C three phase windings, one end of all windings is connected together, and the other end is used to load the voltage. If adding positive voltage in A, the direction of the magnetic field generated by phase A is indicated by the red arrow according to Ampere's law, and the direction of the magnetic field generated by the negative voltage applied to B is indicated by the blue arrow. The direction of the resultant magnetic field of the magnetic fields produced by the two phases windings is indicated by the green arrow, and the resultant magnetic field can be seen as a magnet (drawn on the right). Assuming that the rotor of the motor is also a magnet with only one N pole and one S pole, the motor rotor will tend to align with this magnetic field. Apply voltage to two of the three phases of the stator. We have three options: AB, AC and BC. Considering that each phase voltage has two loading conditions (positive or negative), we have a total of 6 ways of applying the voltage, which means that we can generate six different magnetic field directions. If the six different magnetic field directions are generated in sequence, then the magnet rotor will also be aligned with each magnetic field direction in that sequence, and the rotor will rotate. Please note that the direction of the generated magnetic field is the same as the direction of the phase voltage. For example, after we apply a positive voltage to phase A, the direction of the magnetic field (red arrow) points to the side of the cylindrical winding that is close to the motor shaft along the axis of the A-phase cylindrical winding (The black dot at the beginning of the magnetic field direction arrow in the figure), the voltage is also in this direction.

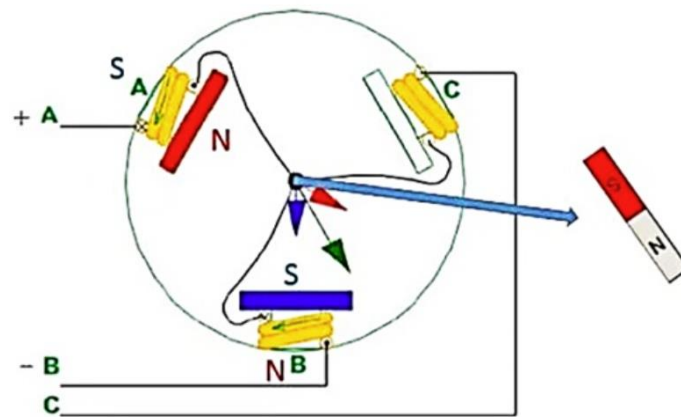


Fig 3. Three-phase motors under magnetic field

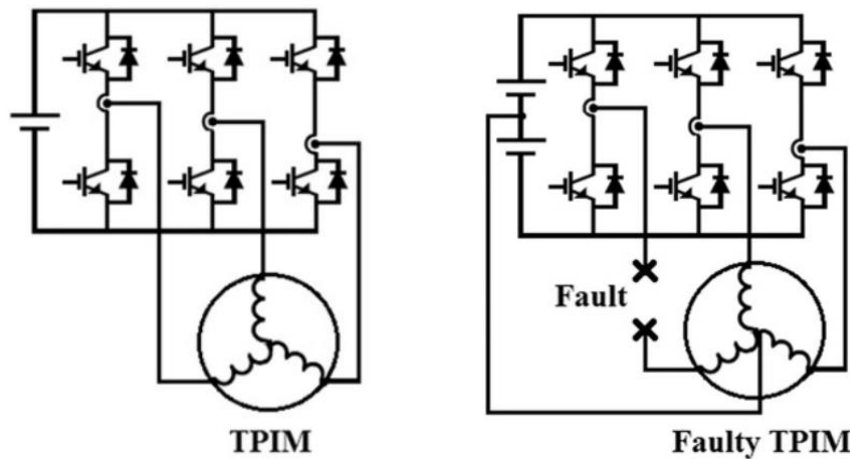


Fig 4(a). Three-phase induction motors

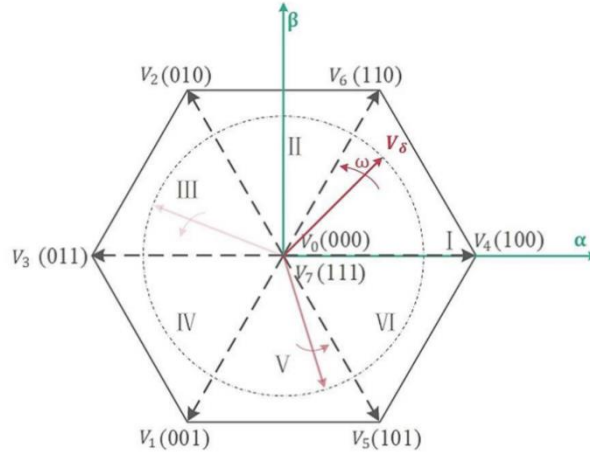
Fig 4(b). Topology used to drive a faulty TPIM

Generally speaking, the research work in the field of post-fault operation of faulty TPIM drive can be divided into two main parts; Inverter topology and control policy. A conventional three-leg voltage source inverter (VSI) powering a healthy TPIM driver is shown in Figure 7.[7] This topology does not have the fault tolerance of Y-connected TPIM drivers during phase-out failures [8]. It is shown in [9] that in order to control the active phase of the star-connected TPIM driver during SOPF, the neutral point (NP) of the motor can be connected to the DC link midpoint. In this study, the topology shown in Figure 1B is used to drive the faulty TPIM. As shown in Figure 8, this topology is the same as the conventional three-legged VSI, except in NP. Under the maximum current and voltage limits of the inverter topology in Figure 1B, the maximum torque will be 1/3 of the rated torque if the excitation current is kept constant. However, if the magnetizing current increases the maximum torque by a factor, then will be the time of maximum torque. If the supply voltage remains unchanged, the magnetizing current during the fault will increase [10] times. It is worth mentioning that this paper presents a general vector control strategy for star-connected TPIM drivers under SOPF with different inverter topologies. In addition to this topology, different inverter topologies, such as redundant branches of the inverter or cascading inverter topologies, can be used for three-phase motors operating under SOPF conditions [11,12]. However, these topologies increase the cost, size, and complexity of the drive system. [7]

## 2. Measurement (Design of SVPWM)

Three-phase voltage

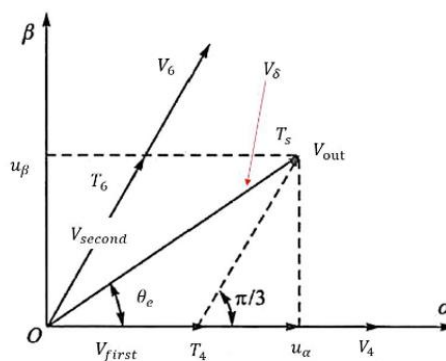
Input the black composite voltage vector in Figure 6 to the SVPWM algorithm  $V_\delta$ , Or enter an equivalent voltage combination that can synthesize this black voltage vector ( $u_a, u_b, u_c$ ). By controlling the states of the 6 switches of the three-phase inverter in Figure 1 it will synthesize the equivalent, PWM-form motor phase voltages, that is, it is reproduced in the form of PWM  $V_\delta$ . This reproduced voltage output can be called  $V_{out}$  (it can be a composite voltage vector or an equivalent combination of voltages). The black composite voltage vector of Figure 6 can be called the reference voltage vector, the trajectory of  $V_\delta$  vertex can be placed in Figure 7



**Fig 5.** Reference voltage vector trace diagram

$V_\delta$  can be in any position, and the amplitude can also be changed. But we only have 6 non-zero vectors and 2 zero vectors, and they are all fixed in magnitude. Let's see how we can reproduce  $V_\delta$  with these 6 non-zero vectors and 2 zero vectors. Every PWM switching cycle  $T_s$ , The SVPWM algorithm can then use the 8 vectors in the figure  $V_0 - V_7$  to produce the  $V_{OUT}$ , the shorter the  $T_s$ , the more the number of combinations for  $V_{OUT}$  in the time of  $V_\delta$  take one rotation.

Now we consume that  $V_\delta$  rotate to the sector 1. We can synthesize it with two adjacent non-zero voltage vectors  $V_4, V_6$  and two zero vectors  $V_0, V_7$  in sector 1, and the combined voltage vector is  $V_{OUT}$ .  $T_4$  is the duration of the voltage vector  $V_4$  in one PWM switching period  $T_s$  (the inverter switching state corresponding to this voltage vector is  $S_a = 1, S_b = 0, S_c = 0$ ). Similarly,  $T_6$  is the duration of the voltage vector  $V_6$  ( $S_a = 1, S_b = 1, S_c = 0$ ) in a  $T_s$ .  $V_{first}$  is in the same direction as the first vector in the counterclockwise direction in each sector (here in the same direction as  $V_4$ ), and  $V_{second}$  is in the same direction as the second vector in the counterclockwise direction in each sector (here in the same direction as  $V_6$ ).  $u_a$  and  $u_b$  are the three-phase voltages  $u_a, u_b, u_c$  of the motor after projecting to the  $\alpha$ - $\beta$  coordinate system shown in Figure 10.



**Fig 6.** Voltage in sectorI

Then according to the equivalent principle of volt-second balance, we have the following relationship ( $T_s$  is small):

$$T_s V_{out} = T_4 V_4 + T_6 V_6 + T_0 V_0 \quad (2)$$

$$T_4 + T_6 + T_0 = T_s \quad (3)$$

In the next step:

$$\frac{T_s}{T_s} V_{out} = \frac{T_4}{T_s} + \frac{T_6}{T_s} + \frac{T_0}{T_s} \quad (4)$$

$$V_{first} = \frac{T_4}{T_s} V_4 \quad (5)$$

$$V_{second} = \frac{T_6}{T_s} V_6 \quad (6)$$

Then we can find from figure 8:

$$|V_{out}| \sin(\theta) = |V_{second}| \sin\left(\frac{1}{3}\pi\right) \quad (7)$$

$$|V_{first}| \sin(\theta) = |V_{second}| \sin\left(\frac{1}{3}\pi - \theta\right) \quad (8)$$

If it is not easy to get the two formulas from Figure 11, you can translate  $V_{second}$  to the position of the dotted line parallel to the right (the right side of the parallelogram). If the dashed line and  $V_{second}$  are parallel, then their same-side interior angles are equal ( $\frac{1}{3}\pi$ ), and their interior angles are also equal ( $\frac{1}{3}\pi - \theta$ ). In addition, the electrical angle is used in Figure 11. In order to generalize the derivation process, we use an arbitrary angle  $\theta$  instead

Combining the above two equations we have the following relation:

$$\frac{V_{out}}{\sin\left(\frac{1}{3}\pi\right)} = \frac{V_{first}}{\sin\left(\frac{1}{3}\pi - \theta\right)} = \frac{V_{second}}{\sin(\theta)} \quad (9)$$

Through the figure4 we can get

$$|V_4| = |V_6| = \frac{2}{3} V_{dc}$$

$$|V_4| = u_m$$

$$T_4 = \sqrt{3} \frac{u_m}{V_{dc}} T_s \sin\left(\frac{1}{3}\pi - \theta\right) \quad (10)$$

$$T_6 = \sqrt{3} \frac{u_m}{V_{dc}} T_s \sin(\theta) \quad (11)$$

$$T_0 = T_7 = \frac{1}{2} (T_s - T_4 - T_6) \quad (12)$$

### 3. Simulation

We decompose the reference voltage into the  $\alpha - \beta$  axis and send it to the SVPWM algorithm module. The SVPWM algorithm can reproduce this reference voltage in PWM form (it can be a three-phase voltage). Figure 28 is the voltages broken down to the  $\alpha - \beta$  axis, when they are not yet in PWM form.

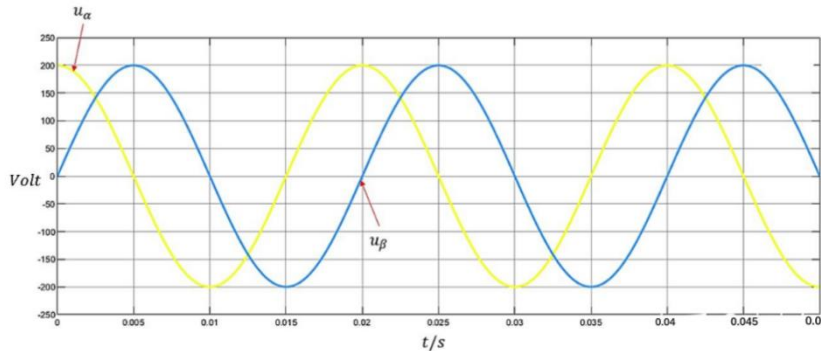
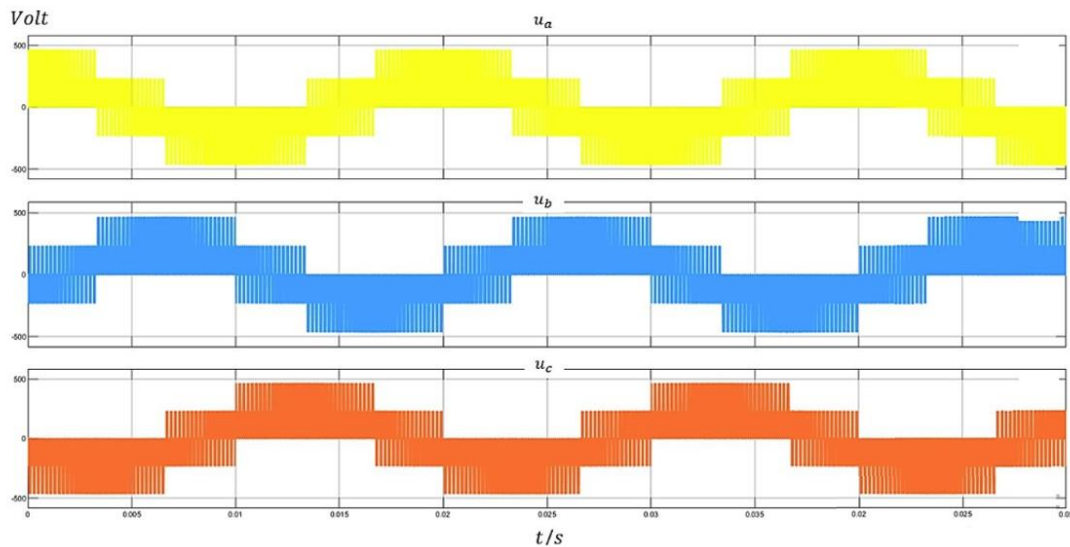


Fig 7. PWM form for  $u_\alpha$  and  $u_\beta$

Then the equivalent three-phase voltage  $u_a, u_b, u_c$  (that is, the three motor phase voltages  $V_{AN}, V_{BN}, V_{CN}$ ) finally reproduced by the SVPWM algorithm is shown in Figure 29. They are all in PWM form. In addition, the DC voltage  $V_{dc}$  in the three-phase inverter (Fig. 1) is 700V.



**Fig 8.** Three phase voltages for motors

## 4. Conclusion

This paper presents the combination between MOSFET circuits (three-phase inverter circuit) and robot arm. To control the robot arm, one of the components for FOC, electronic motors will be connected to this circuit. The PWM wave generated by the specific switching mode composed of the six power switching elements of the three-phase power inverter can make the output current waveform as close to the ideal sinusoidal waveform as possible. The space voltage vector PWM is different from the traditional sinusoidal PWM. It starts from the overall effect of the three-phase output voltage and focuses on how to make the motor obtain an ideal circular flux trajectory. Compared with SPWM, the SVPWM technology has smaller harmonic components of the winding current waveform, which reduces the torque ripple of the motor, makes the rotating magnetic field more circular, and greatly improves the utilization rate of the DC bus voltage, and it is easier to realize digitization.

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